GIVING VOICE THROUGH GAZE

"A Wearable Eye-Controlled AAC Device for Nonverbal Communication"

IMPERIAL

SCAN ME

1 Why It Matters?

WHO? **Cerebral palsy** (CP) is a neurological disorder affecting muscle control and coordination.

17 million — impacted by CP [1]

75% — of children with CP experience speech challenges [2]

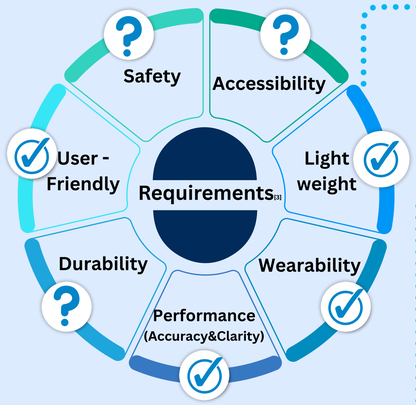
WHAT? **Augmentative and Alternative Communication** (AAC) devices are **tools** that support or replace **speech** to help individuals with limited or no verbal ability communicate .

WHY? While AAC devices exist to help, consultations with specialists revealed that current AACs are often **bulky**, **impractical**, and **stigmatizing**.

2 Our Goal

This project aims to develop a **wearable, user-friendly AAC device** to improve **communication** and **quality of life** for patients and their caregivers.

3 Requirements



How it Works?

The design uses **Near-eye Display** with camera-based **eye tracking** and a projected **circular keyboard**. Users type by focusing on keys, and the system converts input into speech for **hands-free** communication.

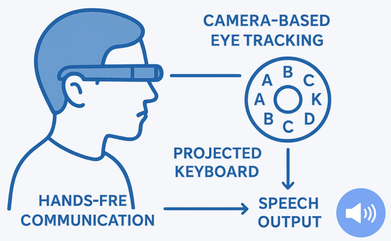
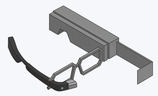


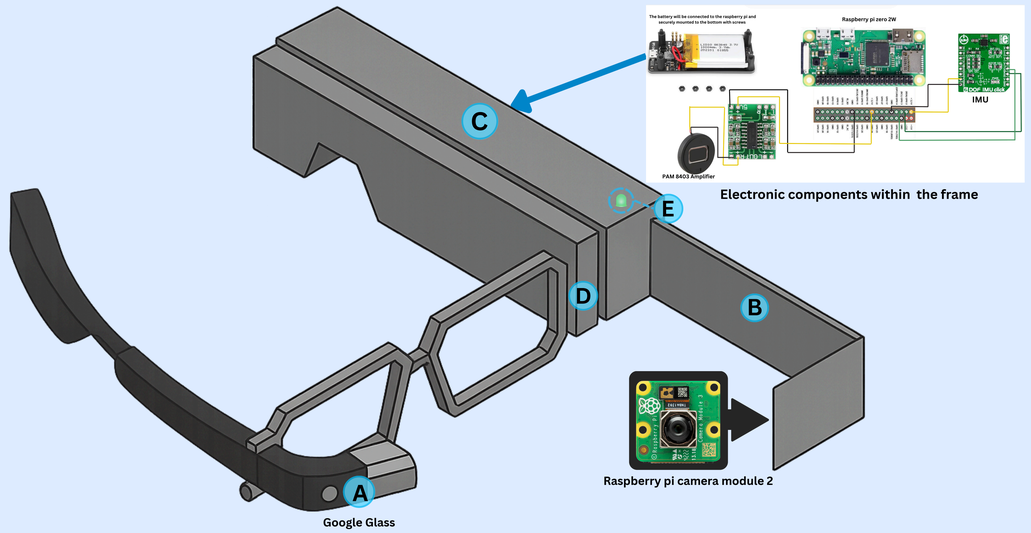
Figure 1. Schematic Representation of Device Functionality.

Frame Design

The frame supports the display lenses **A** and is 3D-printed using **ABS plastic** for **durability** and **resistance** to wear and tear. It includes:

- **B** Extension to hold camera
- **C** Case to hold internal electronics
- **D** Hinge allowing a foldable design





Electronic components within the frame

Figure 2. Final design of the 3D model of our wearable AAC glasses

4 How it's built?

Eye Tracking System

Raspberry Pi Camera Module 2

Real time video input is being fed to the eye-tracking system

1. **Input**
2. **Raspberry Zero 2W** — **Processing**: The video is processed to isolate the pupil and reduce noise
   - **PUPIL ISOLATION**: frames to grayscale and contrast enhancement then ellipse fitting & close-eye estimation
   - **NOISE REDUCTION**: Gaussian Blurring & adaptive thresholding
3. **Gaze Detection**
4. **Output**: Frame position tracked and coordinates are computed

Serial data tuple (x, y, timestamp) sent to Google Glass via Wi-Fi (SPP Profile)

Graphical Interface

Radial Menu Design

Intuitive gaze-based selection

Visual Selection Feedback

Progress indication

Multi-level Selection

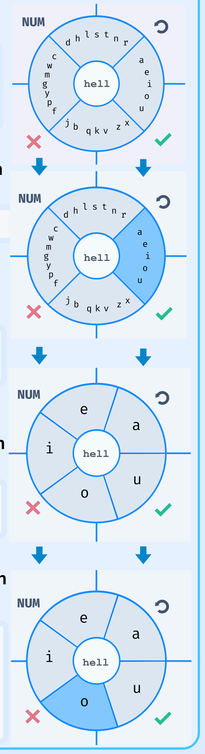
Reducing selection errors

Text-to-Speech Integration

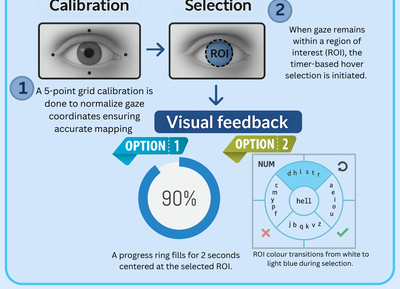
Instant output for communication

Corner Function Buttons

NUM: Number /Character
⟲: Navigation Control
✓: Confirm / Yes
X: Delete / No



Gaze-based selection



Multimodal Interaction

Emergency alert

IMU Sensor MPU-6050
monitors head movement

A help message is triggered when two consecutive left head-tilts are performed

Visual Indicator

Raspberry Pi Zero 2W

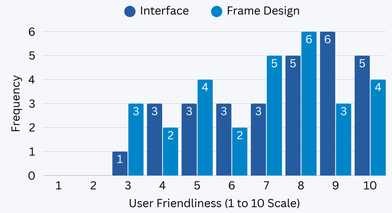
LED light E turns on when the user is speaking

7 Results

Functionality and Performance

- **80%** or greater — **Accuracy** eye-tracking
- **95%** or greater — **Accuracy** IMU trigger
- **7 000-18 000** daily spoken words — **Vocabulary Breadth** via speech-to-text transcription
- **2-5 WPM** — **Speech Output**

Usability

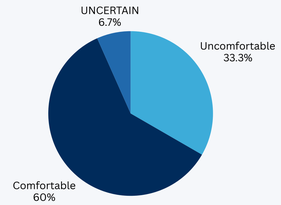


USER-FRIENDLINESS

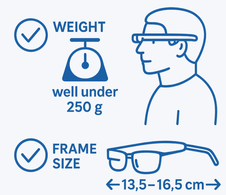
Survey responses rated frame design and interface **user friendliness** predominantly **high**.

WEARABILITY

Survey results indicate that **60%** of respondents would feel **comfortable** wearing the device in **public**.



Size and Weight



Life and Environment

HUMIDITY TEST
functional under 80-90% humidity

DROP TEST
from 1m height

TEMPERATURE TEST
ranging from -15°C to 35°C & surface temperature < 41°C

Safety

EDGE TEST
Sharpness test with PTFE tape

ELECTRICAL TEST
Visual inspection by checking for loose connections and leakage

HEAT TEST
surface remains under 41°C

8 Conclusion

The device was designed with user-centered principles at its core—**wearable**, **lightweight**, and **hands-free**. This design effectively enhances communication for users suffering from **Cerebral Palsy**.

Future Work

ERGONOMICS
sleek and discreet design

COMPATIBILITY
compatible with other AR glasses

PERSONALIZATION
personalized voice-output using ML-trained model

ACCURACY
improve speech precision

LESS EYE STRAIN
using adaptive screen

INTERFACE
refine layout and lexical options

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ACKNOWLEDGEMENTS
We would like to thank the support and guidance of Dr. Ian Radcliffe, Pascal Egan, Luke Thompson, Tariq Malik, Jane Bache and Kate Nelson.

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[1] Graham, H.K., Rosenbaum, P., Paneth, N., Dan, B., Lin, J.-P., Damiano, D.L., Becher, J.G., Gaebler-Spira, D., Colver, A., Reddihough, D.S., Crompton, K.E. and Lieber, R.L. (2016). *Cerebral palsy. Nature Reviews Disease Primers*, [online] 2(1). doi:https://doi.org/10.1038/nrdp.2015.82.
[2] Cerebral Palsy Guidance. (n.d.). *Cerebral Palsy and Speech Therapy*. [online] Available at: https://www.cerebralpalsyguidance.com/cerebral-palsy/treatment/speech-therapy/.
[3]BME Group 3. *BME Group 3 – PSD* [undergraduate project report]. London: Imperial College London; 2025.